

RENYU RAO

Chapel Hill, NC 27516

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Education

University of North Carolina at Chapel Hill

Ph.D. in Statistics and Operations Research; Advisor: Prof. Xiangying (Zoe) Huang

- Research focus: Markov-chain mixing times, random walks on groups.

Aug. 2023 – May 2028 (Expected)

Chapel Hill, NC

Columbia University

M.A. in Statistics; GPA: 3.83/4.33; Statistics Opportunity Scholarship Award (\$50,000)

Aug. 2021 – Dec. 2022

New York, NY

Stony Brook University

B.S. in Mathematics; Double Major in Applied Mathematics & Statistics; GPA: 3.95/4.00

Aug. 2018 – May 2021

Stony Brook, NY

- Major GPA: 4.00/4.00; Summa Cum Laude; Dean's List; Top 20% in Putnam.

Research Experience

Cutoff for Random Walks on Dihedral Groups | Co-author, *arXiv:2510.19942*

Present

- Analyzed continuous-time Markov chains on random Cayley graphs of finite dihedral groups with k i.i.d. random generators; proved high-probability cutoff in nearly the full regime $1 \ll \log k \ll \log |G|$.
- Characterized the phase-dependent cutoff time $t_0(k, G)$ and identified where it differs from Abelian entropic time, linking non-Abelian algebraic structure to total-variation mixing.
- Developed entropy-concentration estimates for auxiliary high-dimensional lattice processes with dependent coordinates; extended high-generator cutoff results to virtually Abelian groups.

Columbia University Department of Statistics

Sep. 2022 – Dec. 2022

Selective Mentee, M.A. Mentored Research; Supervisor: Prof. Bodhisattva Sen

New York, NY

- Studied nonparametric maximum likelihood estimation for Gaussian/location mixture models, connecting mixture-model likelihood geometry with convex-analytic arguments.
- Synthesized academic papers and monographs in weekly reports; established optimizer existence and a finite-support representation via Carathéodory's theorem in likelihood-vector space.

PolyMath Summer Research Program (REU)

Jun. 2021 – Aug. 2021

Participant; Mentor: Prof. Steven Miller, Williams College

Remote

- Collaborated in a five-person research group on Benford's Law for digit-bias diagnostics; simulated precinct-level election profiles under Weibull models in R.
- Implemented chi-square goodness-of-fit tests and sensitivity analyses for digit-bias fraud-detection heuristics on simulated voting data.

Extracurricular

High School Math Competitions

Sep 2016 – June 2018

Participant

New York, NY

- Ranked Top 3% in American Mathematics Competitions(AMC) 12, Invited for American Invitational Mathematics Examination(AIME) and Scored Top 30%.
- Participated in nationwide contests as part of the New York City Math Team, which selects the Top 40 high school students in New York and scored Top 30 team in American Regions Mathematics League (ARML).
- Led Francis Lewis High School Math Team in M3 Modeling Competitions and other major contests like NYSML, NYCIML.

Art of Problem Solving

June 2019 – Aug 2022

Faculty/TA/Grader

New York, NY

- Faculty and TA at Bridge to Enter Advanced Mathematics(BEAM), a nonprofit advanced math summer camp serving 200 middle school students for 3 summers.
- Independently designed and taught two classes in probability and exponential & number theory as a faculty.
- Grading USA Mathematical Talent Search, a national proof-based high school mathematics competition.

New York City Interscholastic Math League

May 2018 – May 2021

Coordinator

New York, NY

- Wrote and reviewed contest problems for 450 students across 35 schools; built a Google Sheets scoring system to automate coach data collection and rankings, helping expand participation by 30%.

Skills

Programming & Tools: R , Python (Basic), Java, $\text{\LaTeX}$, Google Sheets automation, Microsoft Office

Languages: English, Mandarin, Cantonese